

Brain Tumor Classification

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Course: Basic Epidemiology

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Motivation and Objective

- Brain tumors are among the leading causes of mortality.
- Manual diagnosis from MRI is **time-consuming & error-prone**.
- Need for **automated, accurate, and efficient** classification methods.
- Apply **Vision Transformers (ViTs)** for Brain Tumor Classification.
- Classify MRI scans into **4 categories**:

1. Glioma

2. Meningioma

3. Pituitary

4. No Tumor

- Build an **intuitive UI** for real-time prediction.

Dataset and Preprocessing

Dataset

- Source: **Kaggle Brain Tumor Dataset**
- ~7,000 MRI scans across 4 classes
- Data split: **80% training – 20% validation**

Preprocessing

- Resized images to **224×224** (ViT input size).
- Normalized pixel values.
- Applied **data augmentation** (flips, rotations).
- Built a **custom PyTorch DataLoader**.

Vision Transformer (ViT) and Training Setup

- **Patch-based representation** of images (16×16 patches).
- Self-attention mechanism captures **global context**.
- Fine-tuned pre-trained ViT for **4-class classification**.
- Optimizer: **AdamW**
- Regularization: **L2 weight decay**
- Epochs: **25** | Batch size: **32**
- Loss function: **Cross-Entropy Loss**

Results and Comparision

Achieved **Accuracy: 93%**

Performance Metrics:

- Precision: 0.91
- Recall: 0.92
- F1-score: 0.92

CNN Baseline: ~85% accuracy

ViT Fine-tuned: 93% accuracy

ViTs perform better due to **global attention** vs. CNN local filters.

Discussion and Conclusion

Strengths: High accuracy, interpretable attention maps, easy deployment.

Challenges: High computational cost, dataset imbalance.

Future Work:

- Multi-class tumor grading
- Tumor segmentation tasks
- Integration into clinical systems

Fine-tuned **Vision Transformer** achieved **93% accuracy**.

Outperforms CNN-based approaches.

Provides an **intuitive UI** for practical use.

Promising approach for **medical imaging applications**.

References

- Dosovitskiy et al. (2021) – ViT paper
- Vaswani et al. (2017) – Attention is All You Need
- Kaggle Brain Tumor Dataset
- Goodfellow et al., *Deep Learning* (2016)

Thank You!